

Beyond attitudinal ambivalence: effects of belief homogeneity on attitude-intention-behaviour relations

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Abstract

Attitudinal ambivalence extends the traditional unidimensional conceptualization of attitude by acknowledging that people can simultaneously evaluate attitude objects as positive and negative. The present paper argues that this bidimensional view of attitudes may be extended further to take account of multidimensional influences on attitudes using measures of belief homogeneity. Study 1 ($n = 155$) showed that attitudes based on homogeneous belief-sets were significantly more predictive of subsequent behaviour ($\beta = 0.47$, $p < 0.01$) than were attitudes based on heterogeneous belief-sets ($\beta = 0.08$, ns). Study 2 ($n = 136$) manipulated belief homogeneity and found that when beliefs were made heterogeneous, attitudes based on heterogeneous belief-sets were significantly less predictive of behavioural intentions ($\beta = 0.46$, $p < 0.01$) than attitudes based on homogeneous belief-sets ($\beta = 0.84$, $p < 0.01$). Implications for research on attitudinal ambivalence and attribute importance are discussed. Copyright © 2003 John Wiley & Sons, Ltd.

Traditionally, attitudes have been regarded as unidimensional and bipolar, meaning that individuals can hold either a neutral, positive *or* negative attitude towards an object (e.g. a social group, a behaviour). More recently, however, attitudes have been conceptualized as bidimensional constructs whereby individuals can hold positive *and* negative attitudes simultaneously (see Armitage & Conner, in press; Conner & Sparks, 2002; Jonas, Broemer, & Diehl, 2000a; Thompson, Zanna, & Griffin, 1995). The present paper argues that this bidimensional view of attitudes can be further extended to incorporate multidimensional influences on attitude formation, and reports two studies that demonstrate the effects of *belief homogeneity* on attitude-intention-behaviour relations. The following section introduces the concept of attitudinal ambivalence and then outlines how this bidimensional view of attitudes can be extended with reference to the concept of belief homogeneity.

Attitudinal ambivalence occurs when an individual, ‘... is inclined to give it [an attitude object] equivalently strong positive and negative evaluations’ (Thompson et al., 1995, p. 367), and is regarded as extending the unidimensional view of attitudes (that individuals hold either neutral, positive *or* negative attitudes) into a bidimensional model whereby individuals can hold positive *and* negative

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attitudes simultaneously (see Armitage & Conner, in press; Conner & Sparks, 2002; Jonas et al., 2000a; Thompson et al., 1995). Thus, rather than assuming that all positive and negative reactions to attitude objects are integrated into a single unidimensional attitude, the bidimensional view of attitudes recognizes that positive and negative reactions to attitude objects are not always resolved, leaving people feeling ambivalent. Implicit in such research is the idea that ambivalent attitudes are not as well-formed as attitudes that are based on predominantly positive or negative reactions to attitude objects. As such, people with ambivalent attitudes tend to be easier to persuade (Armitage & Conner, 2000, Study 2); to have less persistent attitudes (e.g. M. Conner, submitted; attitudinal ambivalence and the stability of attitudes, 2003); and have attitudes that tend to be less predictive of their intentions and subsequent behaviour (e.g. Armitage & Conner, 2000, Study 1; Conner, Sparks, Povey, James, Shepherd, & Armitage, 2002).

Typically, attitudinal ambivalence is measured by asking participants to rate separately the positive and negative thoughts/feelings they associate with a particular attitude object. For example, Armitage and Conner (2000) presented participants with two items designed to tap the positive and negative poles of attitude; participants were asked to consider only the positive (negative) things about eating a low-fat diet and ignore all the negative (positive) things, before rating how positive (negative) those things were. However, this approach taps only global positive and global negative responses to attitude objects. In the same way that unidimensional measures of attitude do not allow for simultaneous evaluations of objects, perhaps bidimensional measures of attitude mask *multidimensional* influences on attitudes. Thus, one potential problem with the bidimensional view of attitudes is that within each (positive and negative) pole, there are also likely to be inconsistencies. The concept of *belief homogeneity*, on the other hand, acknowledges that attitudes have a multidimensional underpinning. Belief homogeneity taps the extent to which one's beliefs about an attitude object are consistent with one another. For example, whereas attitudinal ambivalence might capture the fact that an individual has both positive and negative thoughts with respect to eating a healthy diet, belief homogeneity potentially captures a more complex view: that eating a healthy diet is somewhat boring, will make you feel very good about yourself, will slightly decrease your risk of coronary heart disease, will greatly reduce your risk of weight gain, but will taste awful. Thus, belief homogeneity extends the bidimensional view of attitudes proposed by the concept of attitudinal ambivalence into a multidimensional view of attitudes.¹ The following section considers how Fishbein's (1963, 1967) expectancy-value model might be utilized to explore multidimensional influences on attitude formation and subsequent effects on attitude-intention-behaviour relations.

Fishbein's (1963) expectancy-value theory is a simple mathematical formula that is held to capture the way in which global attitudes are formed. Ultimately, global attitudes are determined by salient behavioural beliefs: the more positive behavioural beliefs one possesses at any one time (of those beliefs that are salient), the more positive will be one's global attitude (see Fishbein & Ajzen, 1975). In turn, behavioural beliefs are the product of *outcome beliefs* the perceived likelihood of a particular outcome occurring, and *outcome evaluations* the positive or negative evaluation of that outcome. Thus outcomes that are regarded as likely and positive will lead to more positive attitudes, whereas outcomes that are perceived to be unlikely or are evaluated negatively will lead to more negative attitudes.² Operationally, people are asked to rate both the perceived likelihood and their evaluation of a number of possible outcomes occurring; each outcome belief and outcome evaluation is then multiplied and these products are summed to form an index of *behavioural beliefs*, which is used either as a measure of global attitude

¹Note that Jonas, Broemer, and Diehl (2000b, Study 1) report correlations (ranging from $r = -0.20$ to $r = -0.45$) between belief homogeneity and attitudinal ambivalence, suggesting at least some degree of overlap between the two constructs.

²See Trafimow and Finlay (2002) for an interesting discussion of the implications of outcomes rated as both unlikely and evaluated negatively.

(e.g. Parker, Manstead, Stradling, Reason, & Baxter, 1992) or, more commonly, to predict an independent measure of global attitude (for a review see Armitage & Conner, 2001).³

Support for the expectancy-value model of attitudes is widespread and based on studies reporting significant correlations between behavioural belief indices and global attitudes. Several meta-analyses show a large degree of overlap between behavioural belief indices and global attitude. For example, based on 42 independent studies, Armitage and Conner (2001) found a weighted-mean correlation of 0.50 between behavioural beliefs and attitude, while van den Putte (B. van den putte, unpublished manuscript, 1991) reported a mean belief-attitude correlation of 0.53. The present paper explores the idea that global attitudes based on homogeneous belief-sets are better-formed and therefore more predictive of intentions and behaviour than are attitudes based on heterogeneous belief-sets.

To date, just one study has examined the effects of belief homogeneity on one property of attitude strength, namely, attitude stability. Wilson, Hodges, and Pollack (T. D. Wilson, S. D. Hodges, and S. E. Pollack, unpublished manuscript, 1992; cited in Erber, Hodges, & Wilson, 1995) measured belief homogeneity by computing the standard deviation of ten belief terms, to provide an index of '... the variance in each individual's valenced beliefs about the attitude object' (Erber et al., 1995, p. 440). Consistent with their predictions (and their attitudes-as-constructions model), belief homogeneity was a significant predictor of attitude stability. Importantly, Wilson et al. (T. D. Wilson, S. D. Hodges, and S. E. Pollack, unpublished manuscript, 1992) showed that belief homogeneity was more predictive of attitude stability than nine other indices of attitude strength, including attitude accessibility, attitude importance, elaboration and attitude extremity (see Table 17.4 in Erber et al., 1995).

The present paper was designed to extend this research in two main regards. First, Wilson et al.'s (T. D. Wilson, S. D. Hodges, and S. E. Pollack, unpublished manuscript, 1992) findings might be interpreted as reflecting the nature of their attitude object (President Reagan). Because people do not make decisions about politicians *per se*, but about behaviours related to these concepts (e.g. whether to vote for Reagan or not), it is possible that Wilson et al. found greater belief heterogeneity than would be expected if considering one's own behaviour. The second respect in which the present paper extends the work of Wilson et al. (T. D. Wilson, S. D. Hodges, and S. E. Pollack, unpublished manuscript, 1992) is by examining a different property of attitude strength, namely the effects of belief homogeneity on attitude-intention and attitude-behaviour relations, rather than attitude stability (cf. Krosnick & Petty, 1995). Given that homogeneous belief-sets are likely to engender strong attitudes, one would expect that more homogeneous beliefs would augment the attitude-intention and attitude-behaviour relations.

The research reviewed above provides the following rationale for the present paper. Research in the domain of attitudinal ambivalence has tended to show that poorly-formed (i.e. ambivalent) attitudes are less predictive of intentions and behaviours compared with well-formed (i.e. univalent) attitudes (e.g. Armitage & Conner, 2000; Conner et al., 2002), although this research has so far only utilized a bidimensional conceptualization of attitudes. Two studies are reported that test the effects of belief homogeneity on attitude-intention and attitude-behaviour relations.

STUDY 1

Study 1 was designed to test the effects of belief homogeneity on attitude-intention and attitude-behaviour relations. The predictions are that: (a) belief homogeneity will moderate the attitude-intention relationship, such that greater homogeneity will be associated with augmented attitude-intention

³Note that: 'In actuality, although the investigator does perform these computation, people are *not* assumed to do so. We merely propose that attitude formation may be *modeled* in this fashion' (Ajzen & Fishbein, 2000, pp. 7–8).

relations; and (b) belief homogeneity will moderate the attitude-behaviour relationship, such that greater homogeneity will be associated with augmented attitude-behaviour relations.

Method

Participants and Procedure

The present study used 'drinking alcohol' as the target behaviour. The topic was described at the top of the questionnaire, along with instructions as to how to complete the measures. Participants were asked to answer as truthfully as possible and anonymity was assured through the generation of a personal code.

One hundred and fifty five students (82 women, 73 men; mean age = 21) were recruited from the campus of a university in the north of England. One hundred and nine of these participants (70%) were successfully contacted again 1 wk later. The prospective design provided a measure of subsequent behaviour.

Measures

Attitude was measured using a semantic differential scale. Respondents were presented with the sentence stem 'My drinking alcohol in the next week is . . .', which was rated on three 7-point bipolar (−3 to +3) scales *bad-good*, *unfavourable-favourable*, and *negative-positive*. The mean of the three scales was used as the measure of attitude ($\alpha = 0.80$), $M = 0.66$, $SD = 0.99$.

The measure of *belief homogeneity* was based on that used by Wilson et al. (T. D. Wilson, S. D. Hodges, and S. E. Pollack, unpublished manuscript, 1992, see Erber et al., 1995). First, participants were asked to rate five beliefs (generated by a representative sample of the target population) about the consequences of drinking alcohol in terms of both the likelihood that each outcome would occur and the value they attached to each outcome. Both outcome belief and outcome evaluation ratings were made on bipolar (−3 to +3) scales anchored with *unlikely-likely* and *bad-good*, respectively. Second, in accordance with Fishbein's (1963) expectancy-value theory of attitudes, the outcome belief and outcome evaluation terms were multiplied to produce *behavioural beliefs*. The behavioural beliefs were: '... makes me more sociable' ($M = 2.96$, $SD = 3.41$), '... leads to me having poorer physical health' ($M = -2.78$, $SD = 3.69$), '... will result in my becoming dependent on it' ($M = 3.57$, $SD = 5.22$), '... will result in me getting into trouble with authority' ($M = 3.67$, $SD = 4.10$), and '... makes me feel good' ($M = 3.43$, $SD = 3.64$).

Third, consistent with Wilson et al. (T. D. Wilson, S. D. Hodges, and S. E. Pollack, unpublished manuscript, 1992), belief homogeneity was computed within-participants. The standard deviation across the five behavioural belief terms was calculated for each individual and used as a measure of belief homogeneity in subsequent analyses. Fourth, the measure was rescaled so that higher scores indicated greater belief homogeneity ($M = 4.43$, $SD = 1.54$).

Behavioural intention was assessed using three items, each on 7-point bipolar (−3 to +3) scales. The basic construction was: 'I _____ to drink alcohol in the next week *definitely do not-definitely do*', with the words *intend*, *plan* and *want* inserted in the blank space. The mean of these three items produced a reliable scale ($\alpha = 0.82$), $M = 1.35$, $SD = 1.42$.

Behaviour was assessed 1 wk later using three items, each measured on 7-point unipolar (+1 to +7) scales: 'I drank alcohol in the last week *strongly disagree-strongly agree*', 'How often did you drink alcohol in the last week? *never-frequently*', and 'I tried to drink alcohol in the last week *strongly disagree-strongly agree*'. The internal reliability of the scale was good, $\alpha = 0.76$ ($M = 5.41$, $SD = 1.54$).

Table 1. Moderating effects of belief homogeneity on attitude-intention and attitude-behaviour relations (Study 1, drinking alcohol)

	R^2	F	β
1. Predicting intention			
Attitude	0.16	9.70**	0.41**
Belief homogeneity			0.10
Attitude $\times$ belief homogeneity			0.10
2. Predicting behaviour	0.30	7.31**	
Attitude			0.11
Belief homogeneity			0.09
Attitude $\times$ belief homogeneity			0.18*
Behavioural intention			0.40**

* $p < 0.05$; ** $p < 0.01$.

Results

Moderated regression analyses were used to test the hypotheses that homogeneous belief-sets would augment the attitude-intention and attitude-behaviour relations. A moderator effect is shown if the interaction between moderator variable and moderated variable is significant while the independent effects of each are statistically controlled (see Baron & Kenny, 1986). In terms of the predicted moderating effect of belief homogeneity on attitude-intention relations, intention was regressed on attitude, belief homogeneity and the attitude-by-belief homogeneity interaction term (see Table 1).⁴ The three independent variables accounted for 16% of the variance in behavioural intentions, with attitude being the dominant predictor. Contrary to predictions, the attitude $\times$ belief homogeneity interaction term was not a significant predictor of intention, suggesting that belief homogeneity did not moderate the attitude-intention relationship.

A similar procedure was adopted to test the hypothesis that belief homogeneity would moderate the attitude-behaviour relationship (see Table 1). For the purpose of these analyses, behaviour was regressed on behavioural intention in addition to the attitude, belief homogeneity and the attitude $\times$ belief homogeneity interaction term. This controlled for the possibility that there is no direct attitude-behaviour relationship because intentions mediate the effects of attitudes on behaviour (see Fishbein & Ajzen, 1975). Thus controlling for the effects of behavioural intention, the attitude $\times$ belief homogeneity interaction term remained a significant predictor of behaviour indicating that belief homogeneity was a significant moderator of the attitude-behaviour relationship (see Table 1). Importantly, the β -weight associated with the attitude $\times$ belief homogeneity-behaviour relationship was positive, demonstrating that greater homogeneity was associated with an augmented attitude-behaviour relationship. The implication is that attitudes of sufficient strength will predict behaviour directly, over and above the effects of behavioural intention.

The significant moderator effect was decomposed using simple slope analyses (Aiken & West, 1991). Given that high scores on the belief homogeneity scale indicate greater homogeneity, three levels of belief homogeneity were tested: high homogeneity (one standard deviation above the mean on the belief homogeneity scale), medium homogeneity (mean level) and low homogeneity (one standard deviation below the mean, see Figure 1). As the significant interaction showed, when behavioural beliefs were heterogeneous, attitudes were significantly less predictive of behaviour,

⁴In accordance with the recommendations of Aiken and West (1991), the variables involved in the moderator analyses were mean-centred. Mean-centring involves subtracting the mean value for the variable from the raw scores for each participant. This has the effect of making the new mean value equal to zero while retaining the standard deviation of the original score and reduces multicollinearity between the independent variables and their multiplicative composites (see Aiken & West, 1991).

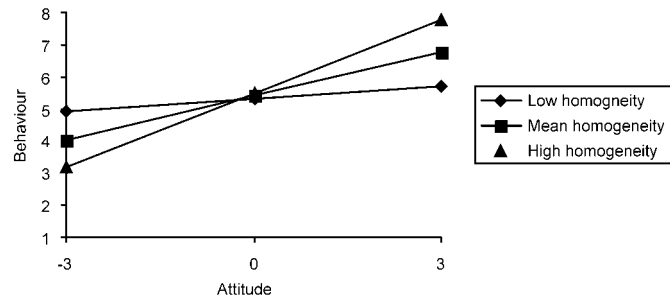


Figure 1. Interaction between attitude and belief homogeneity: simple slopes for attitude at three levels of belief homogeneity (Study 1, drinking alcohol)

$\beta = 0.08$, *ns*, than they were either at mean levels of belief homogeneity, $\beta = 0.27$, $p < 0.01$ or at high levels of belief homogeneity, $\beta = 0.47$, $p < 0.01$. In sum, attitudes based on a homogeneous belief-set were significantly more predictive of behaviour than attitudes based on a heterogeneous belief-set.

Discussion

Consistent with predictions, a homogeneous belief-set produced attitudes that were more predictive of behaviour than did a heterogeneous belief-set. When beliefs were consistent, attitudes were significantly more predictive of behaviour than when beliefs were inconsistent. Of particular interest was the finding that these effects occurred independently of behavioural intention, which has been proposed as a key mediator of attitude-behaviour relations (e.g. Fishbein & Ajzen, 1975). The implication is that attitudes of sufficient strength can predict behaviour directly, over and above the effects of behavioural intentions (cf. Armitage & Conner, 2000). Study 1 extends the findings of Wilson et al. (T. D. Wilson, S. D. Hodges, and S. E. Pollack, unpublished manuscript, 1992) to the domain of self-relevant behaviours and replicates the effects of belief homogeneity on the strength of attitudes using the attitude-behaviour relationship, rather than attitude stability as an indicator of attitude strength (cf. Krosnick & Petty, 1995).

However, contrary to predictions, belief homogeneity did not significantly moderate the attitude-intention relationship. A possible explanation for this finding is that participants were keen to maintain consistency between their attitudes and intentions. Whereas intention was measured on the same (bipolar) scale and at the same time as attitude, behaviour was measured on unipolar scales 1 wk after global attitude was measured. Moreover, this bias might have been enhanced by the instructions to participants, which included an introduction to the attitude object that may have provided participants with an opportunity to reflect on and homogenize their beliefs. It is therefore possible that consistency bias undermined the effects of belief homogeneity on the attitude-intention relationship.

Interestingly, the lack of an effect on the attitude-intention relationship mirrors inconsistencies within the attitudinal ambivalence literature; for example, whereas Armitage and Conner (2000) found that greater ambivalence attenuated the attitude-intention relationship, Conner, Povey, Sparks, James, and Shepherd (2003) found null effects, and Jonas, Diehl, and Brömer (1997) reported that attitudinal ambivalence *augmented* the relationship between attitude and intention. Armitage and Conner's (in press) review of this literature concluded that attitudinal ambivalence was most likely to attenuate the attitude-intention relationship when the attitude object was related to the self and when ambivalence was experimentally manipulated. Given that this conclusion might extend to tests of belief homogeneity, a second study was designed.

STUDY 2

Study 2 was designed both to extend the findings of Study 1 and to address some of the limitations identified in Study 1. The most important innovations were an experimental manipulation of belief homogeneity and an investigation into the effects of belief homogeneity in another behavioural domain, blood donation.

Given that it was suspected that belief homogeneity was unintentionally inflated by the instructions given to participants, it was reasoned that a manipulation involving participants rationalizing versus being distracted from rationalizing their beliefs would be most appropriate. In fact, previous research has shown that the attitude-behaviour relationship is augmented when participants are asked to analyse their reasons for holding a particular attitude (e.g. Fazio, Zanna, & Cooper, 1978; Millar & Tesser, 1986), which would seem to fit the present purpose. Unfortunately, this type of manipulation also affects attitude extremity (see Tesser, Martin, & Mendolia, 1995 for a review), which is empirically and conceptually distinct from belief homogeneity. Empirically, Erber et al.'s (1995) principal components analysis of attitude strength indices showed that attitude extremity and belief homogeneity loaded on separate factors; conceptually, attitude extremity represents an attribute of the attitude itself, whereas belief homogeneity is a feature of the accompanying cognitive structure (cf. Krosnick, Boninger, Chuang, Berent, & Carnot, 1993). The present study adapted the thought-listing methodology to take account of these differences. Given that attitude extremity represents an attribute of the attitude itself, attitude extremity manipulations have typically asked participants to focus on the attitude *per se* (e.g. by listing their reasons for holding that attitude). Therefore, because belief homogeneity is a feature of the accompanying cognitive structure, the belief homogeneity manipulation ought to target people's thoughts surrounding the attitude *object*. Thus, belief homogeneity was manipulated by asking participants to list pros and cons they associated with the attitude object (or an irrelevant attitude object, depending on condition), rather than their reasons for holding the attitude.

The predictions are that: (a) belief homogeneity can be manipulated using an adapted thought-listing task that focuses on the attitude object, rather than the attitude itself; and (b) when beliefs are made heterogeneous, belief homogeneity will moderate the attitude-intention relationship, such that greater homogeneity will be associated with augmented attitude-intention relations, and greater heterogeneity will be associated with an attenuated attitude-intention relationship.

Method

Participants and Procedure

The present study used 'blood donation' as the target behaviour. One hundred and thirty six participants (106 women, 30 men; mean age = 19) were recruited from the campus of a university in the north of England and randomized to either a homogeneous or a heterogeneous condition. None of the questionnaires provided any introduction to the attitude object, nor were they titled. The questionnaires only included instructions as to how to complete the measures.

Given that Study 1 might have inadvertently encouraged participants to reflect on their beliefs, the *homogeneous* condition explicitly asked participants to spend 2 min writing the pros and cons they associated with donating blood. The homogeneous condition was thus designed to increase the level of consistency within participants' blood donation beliefs. In contrast, the *heterogeneous* condition was designed to dissuade participants from thinking about blood donation until they began to complete the questionnaire. Participants randomized to the heterogeneous condition were asked to spend 2 min writing pros and cons they associated with an irrelevant attitude object.

Following the belief homogeneity manipulation, participants in both conditions completed identical measures designed to tap beliefs and attitudes toward blood donation. The measures are described in the following section.

Measures

Attitude was measured using a semantic differential scale. Respondents were presented with the sentence stem, 'My donating blood in the future is . . .', which was rated on two 7-point bipolar (−3 to +3) scales *bad-good*, *negative-positive*. The mean of the two scales was used as the measure of attitude. The scale had adequate internal reliability ($\alpha = 0.69$), $M = 1.94$, $SD = 1.19$.

The measure of *belief homogeneity* was similar to that used in Study 1. Respondents rated eight beliefs (generated by a representative sample of the target population) about the consequences of donating blood in terms of both the likelihood that each outcome would occur and the value they attached to each outcome. Both outcome belief and outcome evaluation ratings were made on bipolar (−3 to +3) scales, anchored by *unlikely-likely* and *bad-good*, respectively. The products of these ratings produced eight behavioural belief terms. The behavioural beliefs were: '...helps others' ($M = 8.40$, $SD = 1.66$), '...gives me a feeling of well-being' ($M = 4.41$, $SD = 4.57$), '...means you get drunk quicker' ($M = -0.27$, $SD = 4.60$), '...makes you feel positive toward others' ($M = 2.66$, $SD = 4.87$), '...makes you feel discomfort (e.g. pain, fainting, nausea)' ($M = -1.96$, $SD = 3.93$), '...means the hassle of having to be careful afterwards' ($M = 0.92$, $SD = 3.15$), '...makes me feel afraid' ($M = 0.57$, $SD = 3.86$), and '...makes me feel vulnerable' ($M = 3.15$, $SD = 4.40$). The standard deviation of the eight behavioural beliefs was calculated for each individual and used as a measure of belief homogeneity. These values were rescaled so that higher scores indicate greater belief homogeneity ($M = 4.83$, $SD = 1.41$).

Behavioural intention was assessed using two items, each on 7-point bipolar (−3 to +3) scales. The basic construction was: 'I _____ to donate blood in the future *definitely do not-definitely do*', with the words *intend* and *want* inserted in the blank space. The mean of these items produced a reliable scale ($\alpha = 0.81$), $M = 1.06$, $SD = 1.66$.

Results

MANOVA showed that the homogeneous and heterogeneous conditions did not differ in terms of age, gender, attitude or behavioural intention, $F(4, 124) = 1.40$, *ns*, indicating that randomization had been successful. None of the univariate F s were significant, most notably the univariate F associated with attitude, $F(1, 127) = 0.06$, *ns*, which shows that the manipulation did not increase attitude extremity. However, the manipulation did affect belief homogeneity. As predicted, the beliefs of participants who were asked to write pros and cons associated with donating blood were significantly more homogeneous ($M = 5.07$) than were the beliefs of participants in the heterogeneous condition ($M = 4.55$), $t(134) = 2.18$, $p < 0.05$.

In order to provide a stringent test of the hypothesis that a more heterogeneous belief-set would attenuate the attitude-intention relationship, a three-way condition $\times$ belief homogeneity $\times$ attitude interaction was tested using multiple regression. This involved testing the belief homogeneity $\times$ attitude interaction separately for each condition and then testing whether the interaction was greater in the heterogeneous condition.⁵ For each condition, intention was regressed on attitude, belief

⁵This particular strategy was adopted because dichotomous moderating variables should be '...treated as different groups' (Baron & Kenny, 1986, p. 1175), and so that the analyses across the two studies would be directly comparable.

Table 2. Moderating effects of belief homogeneity on attitude-intention relations by experimental condition (Study 2, donating blood)

	Homogeneous condition			Heterogeneous condition		
	R^2	F	β	R^2	F	β
Attitude	0.44	15.98**	0.63**	0.43	17.36**	0.65**
Belief homogeneity			0.39**			0.29**
Attitude $\times$ belief homogeneity			0.01			0.22*

* $p < 0.05$; ** $p < 0.01$.

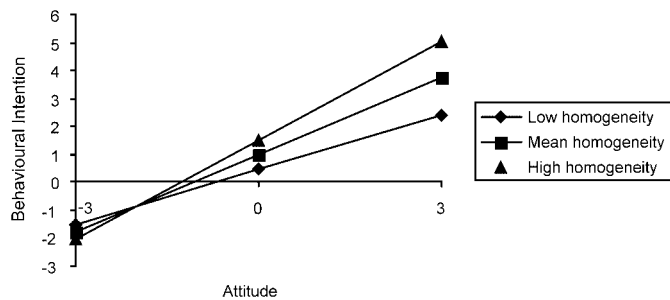


Figure 2. Interaction between attitude and belief homogeneity for heterogeneous condition: simple slopes for attitude at three levels of belief homogeneity (Study 2, blood donation)

homogeneity and the attitude $\times$ belief homogeneity interaction (see Table 2). This procedure ensures that group differences in variance and/or reliability of measures do not produce spurious effects (Aiken & West, 1991; Baron & Kenny, 1986).

In both conditions, the three independent variables accounted for significant proportions of the variance in intention. However, as predicted, the attitude $\times$ belief homogeneity interaction was only significant in the heterogeneous condition, $\beta = 0.22$, $p < 0.05$, indicating that more heterogeneous beliefs were associated with attenuated attitude-intention relations. This significant interaction was decomposed using simple slope analysis (see Figure 2). Consistent with the significant interaction reported in Table 2, at low levels of belief homogeneity, attitudes were significantly less predictive of intentions, $\beta = 0.46$, $p < 0.01$, than they were at either mean levels of belief homogeneity, $\beta = 0.65$, $p < 0.01$ or at high levels of belief homogeneity, $\beta = 0.84$, $p < 0.01$.

The three-way interaction was tested by examining differences in the unstandardized β -weights for the attitude $\times$ belief homogeneity interaction between the two conditions (Edwards, 1984). Consistent with predictions, the attitude $\times$ belief homogeneity interaction was significantly larger in the heterogeneous condition, $t(127) = 1.80$, $p < 0.05$. Thus, increasing belief heterogeneity had the effect of enhancing the effects of belief homogeneity on the attitude-intention relationship.

Discussion

Study 2 demonstrated that it was possible to manipulate belief homogeneity using an adapted thought-listing procedure and that belief homogeneity enhanced the predictive power of attitudes when beliefs were made heterogeneous. The study therefore provides further evidence that attitude extremity and belief homogeneity are distinct and suggests that manipulations designed to affect attitude extremity

should be focused on the attitude *per se* (e.g. by listing reasons for holding that attitude) whereas belief homogeneity manipulations should be targeted at thoughts surrounding the attitude *object* (cf. Erber et al., 1995; Krosnick et al., 1993).

More importantly, Study 2 provides further support for the idea that homogeneous belief-sets enhance the predictive power of attitudes. Whereas existing belief heterogeneity failed to significantly moderate the attitude-intention relationship in Study 1, the heterogeneity manipulation in Study 2 successfully did so. This finding has two main implications. The first is that attitude-intention relations are generally consistent and that belief homogeneity is only likely to affect attitude-intention relations in exceptional circumstances. Consistent with this view, meta-analyses show that attitudes are more closely related to intentions than to any other construct (e.g. Armitage & Conner, 2001). The second implication is that behavioural intentions are not a good proxy for actual behaviour because belief homogeneity seems to affect attitude-intention relations differently from attitude-behaviour relations. In the present study, belief homogeneity only moderated the attitude-intention relationship following an experimental manipulation; in Study 1, existing belief homogeneity failed to moderate the attitude-intention relationship while moderating the attitude-behaviour relationship. Future research should therefore include measures of both behavioural intention and behaviour when considering attitude-behaviour relations.

GENERAL DISCUSSION

The present paper reports two studies that provide evidence for the effects of attitude formation on attitude-intention-behaviour relations. First, controlling for the effects of behavioural intention, attitudes based on heterogeneous belief-sets were significantly less predictive of subsequent behaviour (Study 1). The implication is that sufficiently strong attitudes can supersede the mediating effect of behavioural intentions on the attitude-behaviour relationship. Second, Study 2 extended Study 1 by showing that belief homogeneity could be manipulated and that this manipulation augmented the effects of belief homogeneity on the attitude-intention relationship. Together, these findings show that attitudes constructed on the basis of heterogeneous belief-sets are less predictive than attitudes constructed from homogeneous belief-sets. The following discussion first considers the idea that belief homogeneity extends prior work on attitudinal ambivalence by reflecting a multidimensional underpinning to attitudes before examining the relationship between belief homogeneity and other belief-based measures of attitude strength.

In recent years, researchers have again embraced the topic of attitudinal ambivalence, the idea that individuals can simultaneously evaluate attitude objects with equivalently strong positive and negative evaluations (e.g. Armitage & Conner, *in press*; Conner & Sparks, 2002; Jonas et al., 2000a). One potential problem with the bidimensional view of attitudes is that within each (positive and negative) pole, there are also likely to be inconsistencies. The beliefs that underpin the positive and negative poles are unlikely to be uniform across all individuals. Belief homogeneity, on the other hand, acknowledges that attitudes have a multidimensional underpinning. The present findings show that attitudes based on homogeneous belief-sets have greater predictive validity than attitudes based on an inconsistent database.

However, the effect of belief homogeneity on attitude-intention relations was only apparent when belief homogeneity was manipulated. This finding mirrors some inconsistencies concerning the effects of attitudinal ambivalence on attitude-intention relations; for example, whereas Armitage and Conner (2000) found that greater ambivalence attenuated the attitude-intention relationship, Conner et al. (2003) found null effects, and Jonas et al. (1997) found that attitudinal ambivalence *augmented*

the relationship between attitude and intention. Armitage and Conner's (in press) review concluded that attitudinal ambivalence was most likely to attenuate the attitude-intention relationship when the attitude object was related to the self and when ambivalence was experimentally manipulated. This explains why the moderating effect of belief homogeneity only appeared in the heterogeneous condition of Study 2.

In contrast, Armitage and Conner (in press) found that the effects of attitudinal ambivalence on the attitude-behaviour relationship were relatively clear-cut, reporting only one instance when ambivalence failed to significantly moderate the attitude-behaviour relationship. Consistent with this line of research, Study 1 showed that belief homogeneity significantly moderated the attitude-behaviour relationship such that attitudes based on homogeneous belief-sets were more predictive of subsequent behaviour. These data can also be understood in terms of Wilson and colleagues' (Erber et al., 1995; Wilson & Hodges, 1992) *attitudes-as-constructions* model. Central to the attitudes-as-constructions model is the idea that when attitudes are required (e.g. when making decisions), they are constructed from available information (e.g. accessible memories). One key prediction advanced by the attitudes-as-constructions model is that attitudes based on consistent databases are likely to be more predictive of subsequent behaviour than attitudes based on inconsistent databases (Erber et al., 1995; Wilson & Hodges, 1992). Thus, Study 1 can be interpreted as showing that attitudes based on homogeneous belief-sets were more likely to be reconstructed when actually deciding to perform the behaviour than were attitudes based on heterogeneous belief-sets. Further work is required to determine whether variables other than attitudes drive the behaviour of individuals with heterogeneous belief-sets.

Therefore, belief homogeneity represents an important facet of attitude strength (cf. Erber et al., 1995) that recognizes that attitudes are multidimensional constructs. However, belief homogeneity is not the only belief-based measure of attitude strength and there are parallels between the present findings and research into the effects of *belief importance* on belief-attitude relations (for a review see van der Pligt, de Vries, Manstead, & van Harreveld, 2000). These parallels are explored in the following section.

According to Fishbein's (1963, 1967; Fishbein & Ajzen, 1975) summative model of attitudes, global attitudes are formed from sets of salient behavioural beliefs (outcome belief $\times$ outcome evaluation terms), and some researchers use the summed behavioural beliefs scale to represent global attitude *per se* (e.g. Parker et al., 1992). In spite of this, behavioural belief indices are typically less than perfectly correlated with global attitudes (see Armitage & Conner, 2001; B. van den Putte, unpublished manuscript, 1991). One approach to addressing this issue has been to assess belief importance as a moderator of belief-attitude relations. Belief importance is tapped by asking participants to select the behavioural beliefs that they regard as being most important. For example, Eiser and van der Pligt (1979) presented participants with standard behavioural belief items and asked them to choose the five most important attributes. Eiser and van der Pligt (1979) found that a composite scale based on the five selected attributes correlated 0.86 with a global measure of attitude while the six unselected attributes correlated only 0.44. Similarly, van der Pligt and de Vries (1998) found that the three most important beliefs were strongly correlated ($r = 0.63$) with global attitude, in contrast with the correlation between the 12 less important beliefs and global attitude ($r = 0.15$). Thus, selected important beliefs seem to be more strongly correlated with global attitudes (for a review see van der Pligt et al., 2000).

It seems plausible that importance is attached to beliefs that are consistent with one another, and that the effects of belief importance may be a function of belief homogeneity. It would therefore be valuable to compare the effects of belief importance with those of belief homogeneity in future research. However, it should be noted that the present measure of belief homogeneity has three main advantages over belief importance. First, belief homogeneity is an operative index of attitude strength. In other words, participants are not required to introspect about properties of their own cognitions, and

hence the issue of how participants interpret 'importance' is avoided (cf. Bassili, 1996; Nisbett & Wilson, 1977). Second, research into belief importance has yet to address the question of how many beliefs should be regarded as 'important'. According to van der Pligt et al.'s (2000) review, researchers have asked respondents to nominate between three and five beliefs they regard as 'important'. Third, belief homogeneity is a continuous variable, whereas belief importance is categorical and so more sophisticated multivariate analyses can be utilized to assess the effects of belief homogeneity.

Although the present research demonstrates support for the concept of belief homogeneity, the dearth of research in this area means that further investigation is required. For example, as yet it is unclear what the antecedents of belief homogeneity may be, although it seems likely that people scoring highly on Neuberg and Newsome's (1993) personal need for structure scale would have more homogeneous belief-sets. Similarly, further work on the consequences of belief homogeneity are warranted, most notably to examine whether attitudes based on homogeneous belief-sets are more resistant to persuasion and are used to guide subsequent information processing to a greater extent than attitudes based on heterogeneous belief-sets (cf. Krosnick & Petty, 1995).

In sum, the present research has demonstrated that belief homogeneity moderates the attitude-intention and attitude-behaviour relations, such that greater heterogeneity is associated with attenuated attitude-intention and attitude-behaviour relationships. Further research is required to examine both the multidimensional underpinning of attitudes as well as the relationship between belief importance and belief homogeneity.

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REFERENCES

- Aiken, L. S., & West, S. G. (1991). *Multiple regression: Testing and interpreting interactions*. Newbury Park: Sage.
- Ajzen, I., & Fishbein, M. (2000). Attitudes and the attitude-behavior relation: Reasoned and automatic processes. *European Review of Social Psychology*, 11, 1–33.
- Armitage, C. J., & Conner, M. (2000). Attitudinal ambivalence: A test of three key hypotheses. *Personality and Social Psychology Bulletin*, 26, 1421–1432.
- Armitage, C. J., & Conner, M. (2001). Efficacy of the theory of planned behaviour: A meta-analytic review. *British Journal of Social Psychology*, 40, 471–499.
- Armitage, C. J., & Conner, M. (in press). The effects of attitudinal ambivalence on attitude-intention-behaviour relations. In G. R. Maio, & G. Haddock (Eds.), *Theoretical perspectives on attitudes for the 21st century: The Gregynog symposium*. Hove, England: Psychology Press.
- Baron, R. M., & Kenny, D. A. (1986). The moderator-mediator variable distinction in social psychological research: Conceptual, strategic, and statistical considerations. *Journal of Personality and Social Psychology*, 51, 1173–1182.
- Bassili, J. N. (1996). Meta-judgmental versus operative indexes of psychological attributes: The case of measures of attitude strength. *Journal of Personality and Social Psychology*, 71, 637–653.
- Conner, M., & Sparks, P. (2002). Ambivalence and attitudes. *European Review of Social Psychology*, 12, 37–70.
- Conner, M., Sparks, P., Povey, R., James, R., Shepherd, R., & Armitage, C. J. (2002). Moderator effects of attitudinal ambivalence on attitude-behaviour relationships. *European Journal of Social Psychology*, 32, 705–718.

- Conner, M., Povey, R., Sparks, P., James, R., & Shepherd, R. (2003). Moderating role of attitudinal ambivalence within the theory of planned behaviour. *British Journal of Social Psychology*, 42, 75–94.
- Edwards, A. L. (1984). *An introduction to linear regression and correlation*. New York: W. H. Freeman and Company.
- Eiser, J. R., & van der Pligt, J. (1979). Beliefs and values in the nuclear debate. *Journal of Applied Social Psychology*, 9, 524–536.
- Erber, M. W., Hodges, S. D., & Wilson, T. D. (1995). Attitude strength, attitude stability, and the effects of analyzing reasons. In R. E. Petty, & J. A. Krosnick (Eds.), *Attitude strength: Antecedents and consequences* (pp. 433–454). Mahwah, NJ: Lawrence Erlbaum Associates.
- Fazio, R. H., Zanna, M. P., & Cooper, J. (1978). Direct experience and attitude-behavior consistency: An information processing analysis. *Personality and Social Psychology Bulletin*, 4, 48–52.
- Fishbein, M. (1963). An investigation of the relationship between beliefs about an object and the attitude toward that object. *Human Relations*, 16, 233–240.
- Fishbein, M. (1967). A behavior theory approach to the relations between beliefs about an object and the attitude toward the object. In M. Fishbein (Ed.), *Readings in attitude theory and measurement* (pp. 389–400). New York: Wiley.
- Fishbein, M., & Ajzen, I. (1975). *Belief, attitude, intention and behavior: An introduction to theory and research*. Reading, MA: Addison-Wesley.
- Jonas, K., Diehl, M., & Brömer, P. (1997). Effects of attitudinal ambivalence on information processing and attitude-intention consistency. *Journal of Experimental Social Psychology*, 33, 190–210.
- Jonas, K., Broemer, P., & Diehl, M. (2000a). Attitudinal ambivalence. *European Review of Social Psychology*, 11, 35–74.
- Jonas, K., Broemer, P., & Diehl, M. (2000b). Experienced ambivalence as a moderator of the consistency between attitudes and behaviors. *Zeitschrift für Sozialpsychologie*, 31, 153–165.
- Krosnick, J. A., & Petty, R. E. (1995). Attitude strength: An overview. In R. E. Petty, & J. A. Krosnick (Eds.), *Attitude strength: Antecedents and consequences* (pp. 1–24). Mahwah, NJ: Lawrence Erlbaum Associates.
- Krosnick, J. A., Boninger, D. S., Chuang, Y. C., Berent, M. K., & Carnot, C. G. (1993). Attitude strength: One construct or many related constructs? *Journal of Personality and Social Psychology*, 65, 1132–1151.
- Millar, M. G., & Tesser, A. (1986). Effects of affective and cognitive focus on the attitude-behavior relation. *Journal of Personality and Social Psychology*, 51, 270–276.
- Neuberg, S. L., & Newsome, J. T. (1993). Personal need for structure: Individual differences in the desire for simple structure. *Journal of Personality and Social Psychology*, 65, 113–131.
- Nisbett, R. E., & Wilson, T. D. (1977). Telling more than we can know: Verbal reports on mental processes. *Psychological Review*, 84, 231–259.
- Parker, D., Manstead, A. S. R., Stradling, S. G., Reason, J. T., & Baxter, J. S. (1992). Intention to commit driving violations—An application of the theory of planned behavior. *Journal of Applied Psychology*, 77, 94–101.
- Tesser, A., Martin, L., & Mendolia, M. (1995). The impact of thought on attitude extremity and attitude-behavior consistency. In R. E. Petty, & J. A. Krosnick (Eds.), *Attitude strength: Antecedents and consequences* (pp. 73–92). Mahwah, NJ: Lawrence Erlbaum Associates.
- Thompson, M. M., Zanna, M. P., & Griffin, D. W. (1995). Let's not be indifferent about (attitudinal) ambivalence. In R. E. Petty, & J. A. Krosnick (Eds.), *Attitude strength: Antecedents and consequences* (pp. 361–386). Mahwah, NJ: Lawrence Erlbaum Associates.
- Trafimow, D., & Finlay, K. A. (2002). The prediction of attitudes from beliefs and evaluations: The logic of the double negative. *British Journal of Social Psychology*, 41, 77–86.
- van der Pligt, J., & de Vries, J. (1998). Attribute importance in expectancy-value models of attitudes. *Journal of Applied Social Psychology*, 28, 1340–1355.
- van der Pligt, J., de Vries, N. K., Manstead, A. S. R., & van Harreveld, F. (2000). The importance of being selective: Weighing the role of attribute importance in attitudinal judgment. In M. P. Zanna, (Ed.), *Advances in experimental social psychology* (Vol. 32, pp. 135–200). San Diego, CA: Academic Press.
- Wilson, T. D., & Hodges, S. D. (1992). Attitudes as temporary constructions. In A. Tesser, & L. Martin (Eds.), *The construction of social judgment* (pp. 37–65). Hillsdale, NJ: Lawrence Erlbaum Associates.

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